

COURTFLOW-GS: Online Multi-View Dynamic Gaussian Reconstruction with Fused MLS Tracking

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Abstract

COURTFLOW-GS reconstructs a frame-0 canonical Gaussian scene from a calibrated basketball-camera ring, then tracks dynamic motion with group-aware control points, fused rigid moving-least-squares (MLS), mask/centroid constraints, focused refinement, and periodic keyframes. The final 300-frame diagnostic produces novel-view videos and baked Gaussian checkpoints. All timing below is measured on an RTX 3080 12 GB; RTX PRO 6000 fields are intentionally reserved for replication.

1 Problem and data

The data contains 36 synchronized 4K ring cameras with fixed known calibration. Views $0, 3, \dots, 33$ train the model; the other 24 are held out. Camera poses are never optimized. The raw-capture adapter accepts `cameras/view_XXX.mp4` and `calibration/cameras.txt`, `images.txt`, producing half-resolution pinhole inputs.

2 Methodology

This section specifies the reported configuration so that the method can be recreated without reading the implementation. Let $P_c = K_c[R_c | t_c]$ be a fixed undistorted pinhole camera, with world-to-camera convention $x_c = R_c x + t_c$. Camera poses, intrinsics, and photometric gains are never optimized. RGB is divided by 255. The 12 training cameras are every third camera of the 36-camera ring; all other cameras are held out. Every input is undistorted before segmentation, matching, training, or evaluation.

2.1 Canonical frame-0 reconstruction

Masks and 3D identities. On frame 0, Faster R-CNN detects people (score ≥ 0.5) and SAM2 refines every on-court box; masks smaller than 64 pixels are discarded. Identities are established in 3D rather than by chaining 2D detections. Fit the court plane with RANSAC on full-frame MAST3R triangulations and, when needed, estimate metric scale from a 1.9 m person-height prior. Make a court-aligned grid of 5 cm voxels, 2.4 m high, with a 1 m horizontal margin. Keep a voxel if it projects into the union foreground mask in at least $\max(4, n_{\text{visible}} - 2)$ cameras. Form 26-connected components, reject components with fewer than 30 voxels, height below 1.2 m, or a bottom implausibly above the floor, and split a footprint exceeding 1.1 m into $\lceil \text{maximum footprint extent} / 1.1 \rceil$ floor-plane k-means clusters. Sort components by floor (u, v) to assign deterministic IDs.

Project each component into each camera, splat its projected coarse voxels to a silhouette, and Hungarian-match detector masks to those silhouettes. Accept matches only at $\text{IoU} \geq 0.20$. For an unmatched component, re-prompt SAM2 with its projected box and centroid. Accept this recovery only if its 16-by-16 HSV histogram has Bhattacharyya distance ≤ 0.5 from the median histogram of that component’s accepted masks in other

views. A failed recovery leaves the camera *unlabelled* for that identity—it is not background evidence. Resolve overlapping accepted masks by nearest median component depth. This yields mutually exclusive integer labels L_c (0 is background) and median label depths $d_{c,k}$.

Occlusion-aware visual hull. For identity k , make a 1 cm grid in the coarse-component bounds plus two coarse-voxel margins. Camera c gives a candidate voxel x a negative vote only if x is in its frustum, c has an accepted label for k , $L_c(\pi_c(x)) \neq k$, and no nearer reconstructed surface explains the absence. The absence is explained if either (i) the pixel belongs to another labelled identity j with $d_{c,j} < d_{c,k}$, or (ii) a static depth map obeys $D_c(\pi_c(x)) < z_c(x) - 0.30$ m. Build D_c from full-frame MAST3R points labelled background and colour-consistent in at least two extra views, keeping nearest depth after a 4-pixel square splat; exclude points within 0.5 m of the floor.

Retain x iff it is observed inside k 's label in at least two cameras and has zero unexplained negative votes; then remove voxels below the floor. Therefore a missing silhouette pixel is not forgiven merely because occlusion is possible: it is forgiven only when a measured nearer dynamic or static depth demonstrates an occluder. An unlabelled camera casts no vote. This is the precise meaning of ‘‘occlusion-aware’’ in this method.

Per-instance crop MAST3R points and Gaussian seeds. For every identity and each adjacent/skip-one ring pair $(c, c+1)$ and $(c, c+2)$, use the tight label box, enlarge its side by 35%, enforce a 96-pixel minimum, clip by translation to the image, and resize the square crop to 512×512 . Dilate the source mask by two pixels and query MAST3R reciprocal nearest neighbours only at every second source-mask pixel. Map crop coordinates back with $u = (u_{512} + 0.5)s^{-1} - 0.5 + x_0$ (and analogously for v). Retain a correspondence iff confidence is ≥ 1.5 , both endpoints lie in their identity masks, both depths are positive, maximum reprojection error is below 1.5 pixels, triangulation angle is at least 3° , and the DLT point lies in the one-voxel-dilated hull. It must also be label-consistent and have mean RGB $L_1 < 0.12$ in at least one non-anchor camera.

Those points are accurate surface seeds. Add only surface hull voxels farther than two hull voxels from a seed, thereby filling weak texture without duplicating matched surfaces. Assign each point the median colour across label-consistent self-z-buffer-visible views, remove statistical outliers (20 neighbours, 2 standard deviations), and fuse at 5 mm for people. Background uses full-frame matches outside person hulls dilated by 10 cm and the same colour test, fused at 10 cm. Initialize one normal-oriented, anisotropic degree-1-SH Gaussian per fused point with opacity 0.5, a third-neighbour scale, and normal-axis scale 0.1 times tangent scale. Persons train for 7000 iterations (four cameras/step) and background for 10000 (two cameras/step), using

$$\mathcal{L}_{\text{image}} = \|I - \hat{I}\|_1 + 0.2\text{D-SSIM} + 0.1\mathcal{L}_{\text{alpha}} + 0.05\mathcal{L}_{\text{depth}}.$$

Depth weight decays to zero halfway through training. Densification starts at iteration 500 and runs every 100 iterations until halfway; prune opacity below 0.01 and splats outside a 3-pixel-dilated mask in two views.

2.2 Group-aware control-point MLS

Every dynamic Gaussian belongs to exactly one person (the ball, when enabled, is a separate group); background is group 0. Farthest-point sample 1024 controls per person, 16 for the ball, and 1000 for background. Each Gaussian is bound once to its $K = 8$ nearest controls from the *same* group. Its fixed radial-basis weights are

$$w_{ij} = \frac{\exp(-\|x_i - p_j\|^2 / (2\sigma_j^2))}{\sum_{\ell=1}^8 \exp(-\|x_i - p_\ell\|^2 / (2\sigma_\ell^2))},$$

where σ_j is the mean distance from control j to its three nearest controls. The six nearest same-group controls form the ARAP graph. Thus a player's splats cannot be moved by any other player or by background controls.

At frame f , warm-start control offsets from the previous control velocity. Optimize a shared translation T_k per movable group and local control offsets δ_j for 50 Adam iterations, using four cameras/iteration and learning

rates 0.020 m and 0.003 m respectively. Let $t_j = T_{g(j)} + \delta_j$, $p'_j = p_j + t_j$, and retain the fixed weights above. For Gaussian i ,

$$\bar{p}_i = \sum_j w_{ij} p_j, \quad \bar{q}_i = \sum_j w_{ij} p'_j, \quad M_i = \sum_j w_{ij} (p_j - \bar{p}_i)(p'_j - \bar{q}_i)^T.$$

With $M_i = USV^T$, set $R_i = VU^T$, flipping the last column of V if $\det(VU^T) < 0$. The baked position is $x'_i = R_i(x_i - \bar{p}_i) + \bar{q}_i$; orientation and degree-1 SH are also rotated by R_i . The fused CUDA operation evaluates this transform and its backward pass. The reported run does not blend separately optimized control rotations.

Every fifth frame, render per-instance weights, prompt SAM2 with each rendered box and centroid, and resolve overlap by highest rendered instance weight. Triangulate each mask centroid by area-square-root-weighted least-squares ray intersection. Use the resulting displacement to initialize T_k , and apply a Huber anchor of scale 0.3 m and weight 0.1 when enough views observe the identity. On intervening frames, no mask or anchor is fabricated. The deformation objective is

$$\mathcal{L} = \mathcal{L}_{\text{image}} + 0.3\mathcal{L}_{\text{BCE}}^{\text{instance}} + \mathcal{L}_{\text{ARAP}} + 0.01\mathcal{L}_{\text{accel}} + \mathcal{L}_{\text{anchor}},$$

where ARAP averages $\|(p'_a - p'_b) - R_a(p_a - p_b)\|^2 / \|p_a - p_b\|^2$ over graph edges and acceleration is normalized second-order control displacement. During deformation, background controls are frozen and the dynamic raster is depth-composited over a cached background.

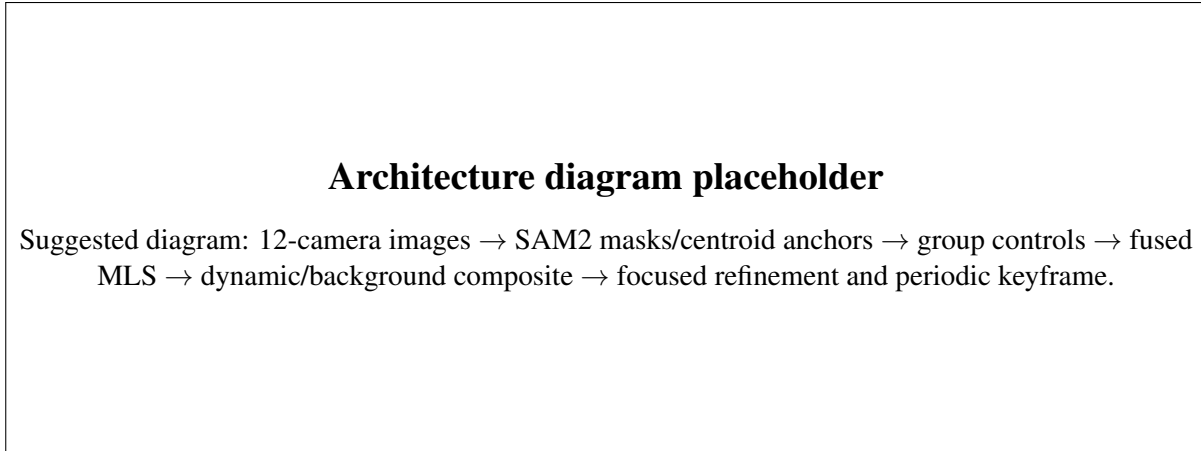


Figure 1: Reserved space for a final CourtFlow-GS system diagram.

2.3 Focused refinement

Bake Gaussian and control state forward after every deformation. At a non-keyframe, project all dynamic Gaussians into every training camera, form their union box with 48 pixels of padding, snap it outward to the 16-pixel gsplat tile grid, shift the principal point by the crop origin, and run 100 refinement iterations on two crops/step. Tile alignment makes crop rasterization pixel-identical to the corresponding full render. This phase has no densification or pruning, so control bindings remain stable. Every tenth frame instead runs 500 full-image iterations, permits densification from iteration 50 every 50 steps until 60% of the schedule, caps growth at 5% per keyframe, rebinds each Gaussian to eight nearest same-group controls, and refreshes the background cache. Held-out cameras are never used in a loss.

2.4 Pseudocode

Algorithm 1 CourtFlow-GS online reconstruction

Require: Frame-0 images, synchronized training videos, fixed calibration

```

1:  $(state, controls) \leftarrow \text{BUILD CANONICAL SCENE}(frame_0, calibration)$ 
2: for each  $f$  in the video sequence do
3:    $images \leftarrow \text{DECODE UNDISTORT NORMALIZE}(trainingVideos, f)$ 
4:   if  $f \bmod 5 = 0$  then
5:      $masks \leftarrow \text{SAM2 REPROMPT}(images, state.dynamic)$ 
6:      $anchors \leftarrow \text{TRIANGULATE MASK CENTROIDS}(masks)$ 
7:   else
8:      $masks, anchors \leftarrow \emptyset, \emptyset$ 
9:   end if
10:   $offsets \leftarrow \text{OPTIMIZE CONTROLS}(L_1 + D\text{-SSIM}, masks, anchors, ARAP)$ 
11:   $state \leftarrow \text{BAKE FUSED RIGID MLS}(state, controls, offsets)$ 
12:  if  $f \bmod 10 \neq 0$  then
13:     $state \leftarrow \text{REFINE DYNAMIC CROPS}(state, images, 100)$ 
14:  else
15:     $state \leftarrow \text{REFINE FULL IMAGES}(state, images, 500)$ 
16:     $controls \leftarrow \text{REBIND}(state, controls); \text{CACHE BACKGROUND}(state)$ 
17:  end if
18:   $\text{SAVE DIAGNOSTICS}(state)$ 
19: end for

```

3 Results

Table 1: Quality summary. Temporal held-out samples coincide with keyframes.

Method	Full PSNR	Person PSNR	Person IoU	Measurement
Global 3DGS baseline	17.50	–	–	frame-0 held-out
Stage A canonical	19.32	18.34	0.788	frame-0 held-out
Stage B v1	18.29	15.00	0.70→0.30	keyframe held-out mean
Stage B v2 (old)	18.87	19.62	0.71→0.39	keyframe held-out mean
Focused refinement, frames 1–20	22.42	26.76	0.746	training-view mean
Final 300-frame diagnostic	20.07	22.44	–	held-out mean



Ground truth, view 13, frame 200



Earlier Stage B v2



CourtFlow-GS diagnostic

Figure 2: Same-camera comparison at held-out view 13, frame 200. Replace or add comparison rows as desired.

4 RTX 3080 runtime

Table 2: Runtime.

Component	RTX 3080 12 GB	RTX PRO 6000
MLS deformation/control optimization	1.95 s/frame	[measure on RTX PRO 6000]
Focused crop refinement	3.41 s/non-keyframe	[measure on RTX PRO 6000]
500-step full keyframe	23.16 s/keyframe	[measure on RTX PRO 6000]
Mask refresh, amortized	0.33 s/frame	[measure on RTX PRO 6000]
End-to-end diagnostic	7.61 s/frame (0.131 FPS)	[measure on RTX PRO 6000]
GPU SH colour + baked raster, 937×527	4.36 ms (229 FPS)	[measure on RTX PRO 6000]

The fused MLS/SVD path is a small fraction of tracking time; rasterization forward/backward dominates optimization. Baked-state GPU rendering is fast, while Python-to-ffmpeg transfer limits the current encoded-video exporter.

5 Limitations and next steps

Crowded late-game plays cause independent per-camera SAM2 identities to swap or disappear. In the final diagnostic, training IoU falls from 0.789 near the start to 0.001 at frame 299; held-out person PSNR falls from 23.86 to 20.59 dB. The highest-priority improvement is detector-led, multi-view identity association at keyframes. Additional work includes checkpoint consolidation or LOD for the 2.01M-Gaussian final state, unbiased inter-keyframe held-out evaluation, and direct GPU-to-NVENC output.